OPTIMIZED TRAFFIC LIGHT SYSTEM USING YOLO and DEEPSORT

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***Abstract*—** In the latest development Traffic management systems are increasingly important to deal with the inefficiencies caused by constant waiting times of road traffic to improve urban mobility. The project focuses on developing a dynamic solution that uses object detection, tracking, and speed estimation to optimize traffic. Using YOLOv8 and DEEPSORT, drivers will be automatically detected and tracked. This facilitates more effective signal control when crossing the road. The system assigns a unique ID to each driver it detects. and follow with demarcation boxes and style lines. This improves the visual overview. Additionally, a vehicle counting mechanism is integrated by defining directional boundaries (north, south, east, west) and counting vehicles as they cross these specified lines. Equipped with a flashing indicator of vehicle entry and exit into a manually defined area. Speed estimation is performed using the Euclidean distance formula, which provides important information for traffic analysis. Future improvements could include the ability to detect pressurized vehicles such as ambulances and fire trucks. So that the system can prioritize these vehicles during crises and dynamically adjust traffic lights. This reduces waiting times and improves preparedness. This project offers a comprehensive approach to optimize traffic flow and improve overall road performance using real-time analytics and object tracking.

Keywords: Object Detection, Object Tracking, YOLOv8, Deep SORT, Vehicle Counting, Computer Vision, Deep Learning, OpenCV

I. INTRODUCTION

India's development has brought a steady increase in vehicle ownership, influenced by improving economic conditions, social advancement, and changing cultural preferences. Therefore, the increase in the amount of vehicles on the road which causes traffic congestion. This problem results in traffic jams. Long period of use Excessive fuel consumption and negative impacts on the environment Traffic management is important. And intelligent traffic management systems using real-time data analysis and dynamic control have become an important solution. An important aspect of intelligent traffic management is the ability to instantly track drivers and understand their movement patterns. with accurate traffic monitoring You will gain valuable insights into vehicle density, speed, and direction. This data service allows dynamic traffic schedule adjustment. Optimizes traffic flow, reduces queues, and improves overall efficiency of the traffic system.

At the core of this intelligence is YOLOv8, a doping model for real-time object recognition. YOLOv8 is known for its speed and cunning. This method is ideal for identifying drivers in video streams near traffic cameras. This model can detect driving in various traffic conditions. quickly and immediately, this includes complex situations with obstacles, dirt, and varying lighting conditions. By using YOLOv8's high-altitude object detection, the system can effectively identify vehicles. and lay the foundation for further analysis. Updated vehicle identification is necessary to ensure that tracking algorithms can continuously track vehicle movement if the vehicle is at an intersection or on the road. After vehicle traffic is detected using YOLOv8, for effective tracking, The DEEP SORT (Simple Online and Realtime Tracking with a Deep Association Metric) algorithm is used . Deep SORT determines the ID. specific to each detected driver. This is so that the system can track individual drivers over time. Even if the driver is temporarily blocked or removed from the camera's view, Deep SORT can continue to track the driver when it pops up. and maintain tracking ID consistency. The power of Deep SORT lies in its ability to match detected object (vehicle) images one at a time. and ensure that each vehicle is consistently tracked as it moves through traffic. This tracking information is important for understanding vehicle movement patterns. including speed and direction This can be used to dynamically adjust traffic times.

The combination of YOLOv8's two object detection resources and two tracking resources forms Deep SORT based on the improved traffic light system. By analyzing the actual movement of two cars. The system can then make intelligent decisions about the timing of both traffic lights. For example, the duration of a green signal may be extended at high speeds. or prioritize certain errors based on the driver's speed and movement. Contribute to minimizing ongoing waiting times. This improved system has the potential to reduce overload. Improve traffic flow and reduce greenhouse gas emissions by adjusting traffic plans to local traffic conditions. The system's efficiency in managing traffic helps make transitions during crises smoother. Reduce the use of rockets and pollution.

II. RELATED WORK

Traffic signal optimization has evolved significantly with the development of various models aimed at improving urban mobility. Traditional approaches, such as the static time interval model, operate on fixed time cycles regardless of real-time traffic conditions. While cost-effective and simple to implement, this model often results in unnecessary delays and increased fuel consumption, particularly in fluctuating traffic patterns 【1】.

In contrast, the density-based traffic signal control model offers a more adaptive solution by utilizing sensors to measure vehicle density approaching intersections. This method effectively reduces wait times and congestion, although its reliance on sensor accuracy can be problematic, particularly with non-standard vehicles like bicycles or pedestrians【7】. Adaptive Traffic Signal Control (ATSC) represents a significant advancement, which is dynamically changing the signal timings by utilizing real-time data and AI algorithms. This model predicts future traffic flow based on historical data and current patterns, enabling more efficient traffic management. However, the complexity and infrastructure costs associated with ATSC pose significant challenges【2】. Emerging technologies such as Vehicle-to-Infrastructure (V2I) communication facilitate direct communication

between vehicles and traffic signals. This model can enhance traffic flow by allowing signals to respond dynamically to real-time vehicle data. However, the widespread implementation of V2I still faces challenges related to technology adoption and collaboration among stakeholders［6］

The literature highlights several drawbacks of existing systems. Static models can not adapt to dynamic conditions , leading to congestion and inefficient traffic flow【5】. Density-based systems may struggle with accuracy and cost due to sensor infrastructure requirements【8】. ATSC, while efficient, requires significant investment and continuous maintenance【9】. Furthermore, V2I technology is still nascent, necessitating collaboration between various entities for effective deployment【12】. Research indicates that integrating deep learning and computer vision techniques, such as YOLOv8 for object detection and Deep SORT for tracking, can optimize traffic signals more effectively. YOLOv8 improves upon its predecessors by enhancing speed and accuracy, making it suitable for real-time traffic applications【13】. Studies show that the combination of YOLOv8 and Deep SORT can significantly improve traffic management, with real-time adjustments leading to reduced waiting times and enhanced safety at intersections【14】.

In this research, frame differencing techniques are employed to detect and track moving objects in video footage. Frame differencing identifies motion by subtracting the current frame from a reference or previous frame, demonstrating effectiveness in various situations. Subsequently, a Kalman filter is utilized to track the detected objects across frames, creating their trajectories. This combination facilitates efficient surveillance capabilities, enabling users to receive notifications via SMS, email, or online streaming in the future【11】. Frame differencing, coupled with Kalman filter tracking, provides a robust system for recognizing and monitoring moving objects, essential for surveillance applications【10】.

The integration of advanced speed estimation techniques further enhances the efficacy of traffic signal optimization systems. Accurate speed estimation allows for dynamic signal adjustments, leading to better traffic flow and improved safety【3】. However, challenges remain, including the need for substantial computational resources and the impact of adverse weather conditions on detection accuracy. In addition to frame differencing and Kalman filter tracking, other techniques can enhance object detection and tracking systems. Background subtraction, which models static scene elements, complements frame differencing for improved handling of lighting changes or shadows. Optical flow can also track pixel intensity shifts, providing more precise motion detection, especially for slower or smaller objects. Techniques like convolutional neural networks (CNNs) in machine learning, can further enhance object recognition and differentiate between various types of moving objects, enhancing system robustness. To address poor visibility in adverse weather, integrating thermal imaging or radar technology can boost detection accuracy. Cloud computing and IoT infrastructure enable real-time video processing and quicker alerts, making these systems more efficient for surveillance and traffic monitoring applications.

III. METHODOLOGY

The optimized traffic light system using computervision which has been proposed leverages YOLOV8 and DEEPSORT. This combined procedure makes sure that the model can able to dynamically adjust traffic signal timings to improve traffic flow and reduce congestion. This study consists of three stages which are data collection, data cleaning and preprocessing, and finally model implementation, each step used to handle the traffic congestion.

A. *Data Collection*

The first step in collecting data is to create a detection model that can be achieved by collecting images of the vehicle. We collect data from multiple platforms to increase the model's ability to detect different types of vehicles. The main dataset is collected from Carvana, which provides a wide range of vehicle images. Videos are also sourced from Shutterstock, which allows us to incorporate dynamic elements into training our models. The dataset we collected is a custom build published on Kaggle containing approximately 10,000 images. This comprehensive dataset allows the model to learn from a wide range of vehicle types and characteristics. This ensures better recognition accuracy. We aim to improve the model's performance in real-world use using a diverse collection of images and videos.

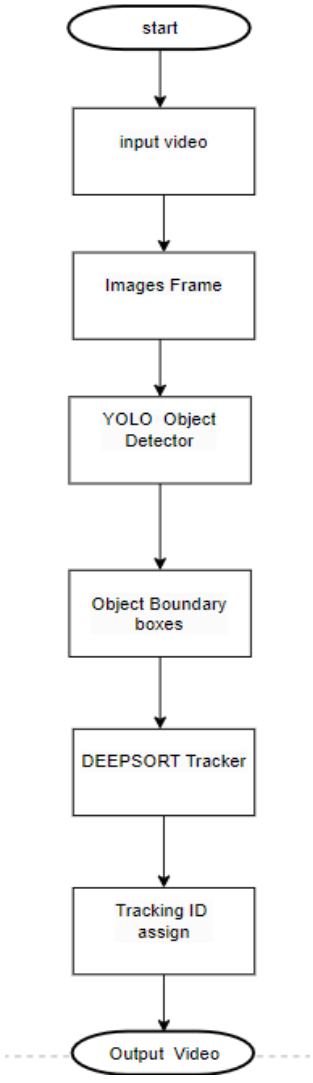


Fig. 1. Methodology flowchart

B. *Data Cleaning and Integration*

The cleaning process consists of three main steps. First, the dataset is carefully filtered to identify and remove corrupted images and video frames that do not meet the quality criteria. Second, the problem of

missing or incomplete data points is obtained. Corrected Items with too many null values or improper formatting are eliminated to maintain data set consistency. For the case where there is the least missing value Statistical imputation techniques were used. It uses the mean and median of related properties to fill in data gaps. in the last step Image standardization was performed to ensure consistent input characteristics across the dataset. This involves normalizing all images to a consistent dimension and color representation. A Min-Max scale is used to transform pixel values to the range [0, 1], which prevents feature dominance during the training process. and promote more efficient learning models.

C. *Training YOLO model*

In this system, YOLOv8 separates itself from object detection based on its ability to segment specific datasets and leverages transfer enhancement techniques. DEEPSORT was chosen for object tracking because of its capabilities. Special to maintaining the identity of objects across frames, YOLOv8 detects objects, while DEEPSORT assigns a unique ID to each detected object. It is what allows frame tracking. Continuous movement analysis post-processing also includes filtering out duplicate detections and low-confidence components using Isso non-maximum suppression, which contributes to a guaranteeing cleaner and more accurate tracking results. By using these techniques, the system can therefore effectively control real-time movement. Provide valuable traffic management information and improve overall operational efficiency. This robust structure can also be adapted to a variety of application areas. Including safety management and urban planning. and demonstrate the scope and potential of cross-sectoral impacts.

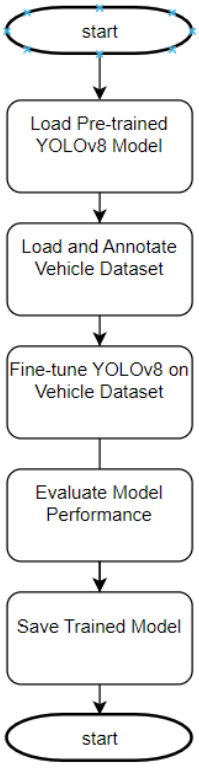


Fig. 2. Yolo Model Training

D. *Object Detection and Tracking Module*

The Object Detection and Tracking Module integrates YOLOv8 for real-time detection of vehicles, pedestrians, and other relevant objects. YOLOv8 provides high-speed, accurate detection, while Deep SORT tracks detected objects across frames. Deep SORT ensures that each object maintains a unique identity, allowing for consistent tracking over time. This integration allows for the detection and tracking of multiple objects in dynamic, complex traffic environments, crucial for real-time traffic management and signal optimization applications.

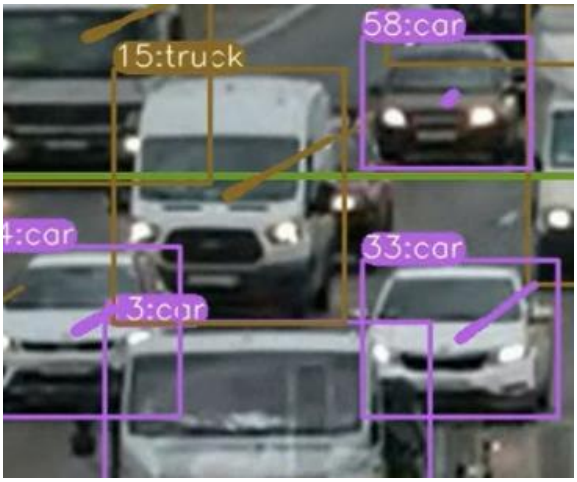


Fig. 3. Vehicle Image detection

E. *Speed Estimation Module*

The speed calculation module calculates the vehicle's speed using the Euclidean distance formula. It measures the straight-line distance between two points. It can also be used to measure distances in Manhattan. It calculates the distance between two points by summing the absolute difference of their Cartesian coordinates. The distance left by the driver is determined between consecutive images and divided by the time traveled. The driver's position is tracked by Deep SORT, which ensures instantaneous object identification and movement tracking across multiple frames. This speed estimation is useful for analyzing traffic flow and overload levels. and provides important information for the following traffic light control module to adjust signal timing. with instant estimation of vehicle speed The system will help optimize traffic management, reduce queues and improve overall efficiency in the face of future traffic situations.

$$\boldsymbol{d(p,q) = \sum_{i=1}^{n} |p_i - q_i|}$$

F. *Traffic Signal Control Module*

The traffic light control module dynamically optimizes signal timing by analyzing data from sensors, tracking and speed estimation. It adjusts the signal phase according to real-time driving instructions. Speed estimation and general traffic patterns Advanced algorithms such as adaptive signal control Cycle length optimization and prioritizing data transmission It is used to improve crossing efficiency. By responding to different traffic conditions, the system will reduce traffic, reduce delays, and improve overall traffic flow. and guarantee better traffic management and traffic safety in reality. This not only improves travel times. It also optimizes fuel consumption and reduces emissions. The system also incorporates predictive traffic modeling to predict future congestion and adjust

IV. *RESULTS AND DISCUSSION*

A custom dataset containing 10,000 images was created using a network scraping technique that captures multiple vehicle types, such as cars, trucks, and bicycles. Additionally, some images were excluded from the existing dataset, such as detection. COCO images for variation and target range The dataset is divided into 70 % for modification and 30 % for validation. It follows standard practices for developing machine learning models. The review set includes selections from publicly available sources. This balanced dataset is designed to facilitate the development and evaluation of robust object detection and tracking models

The adaptive traffic light control system uses YOLOv8 to display driving directions in real time. It is used to dynamically adjust signal times based on traffic density. However, formal quantitative experiments are not alone required. But the system is designed to reduce drivers' waiting times. and improve driving through normal road intersections by changing the timing of green lights to suit traffic speeds. In the case of low traffic speed, the green phase will be shortened to avoid unnecessary delays. While you are in high density traffic The system will signal a green light to block traffic.

Expected benefits from this change include reduced traffic congestion. Minimal delay and better urgent management Future work will focus on rigorous experimental validation to evaluate improvements in computations such as average waiting time, throughput, and accuracy. Compared with the traditional fixed signal control system. This system employs predictive analysis of traffic trends and integrates IoT sensors to boost efficiency while prioritizing pedestrian and public transport routes, ultimately improving response times and traffic safety in urban areas. These functions are essential for understanding system performance in real-world environments. Furthermore, the integration of machine learning algorithms can facilitate adaptive signal control, allowing the system to learn and adjust to traffic patterns over time. The goal is to create a seamless urban mobility experience that benefits all road users. By continuously refining the system, we can pave the way for smarter, safer cities.

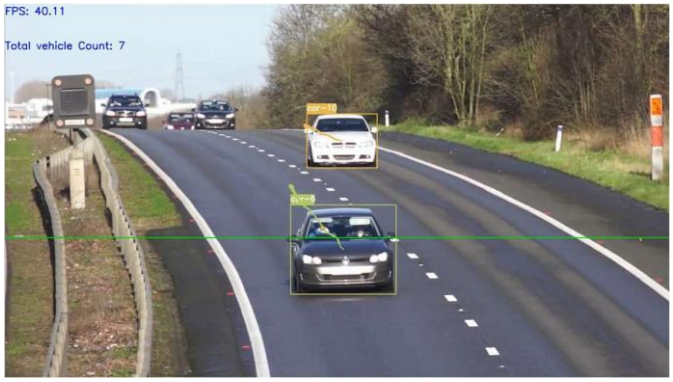


Fig. 4. Yolov8 Car Detection

Table-I Qualitative analysis of the proposed system

class	Images	Instances	Precision	Recall
car	619	1963	0.882	0.918
motorcycle	619	59	0.859	0.678
truck	619	25	0.673	0.56
van	619	38	0.813	0.658

It shows a traffic light system which uses YOLOv8 for object detection and DEEPSORT for tracking. Demonstrates outstanding performance in many types of driving. The system was validated on a dataset of 619 images with 2,185 vehicle traffic events, including cars. Motorcycles, Trucks, and Trucks Evaluation is based on precision, recall, and mean Precision (mAP).

This model shows outstanding speed in vehicle detection. And with an accuracy of 0.882 and a recall of 0.918, this high performance improves the system's performance in quickly identifying and locating vehicles. Especially in urban traffic situations, however, motorcycle detection also produced excellent results with an accuracy of 0.859, indicating significant detection. For heavy vehicles such as trucks and trucks, this model still maintains strong performance. In particular, the vehicle has an excellent visibility of 0.813, demonstrating the model's ability to detect various types of non-traffic movement.

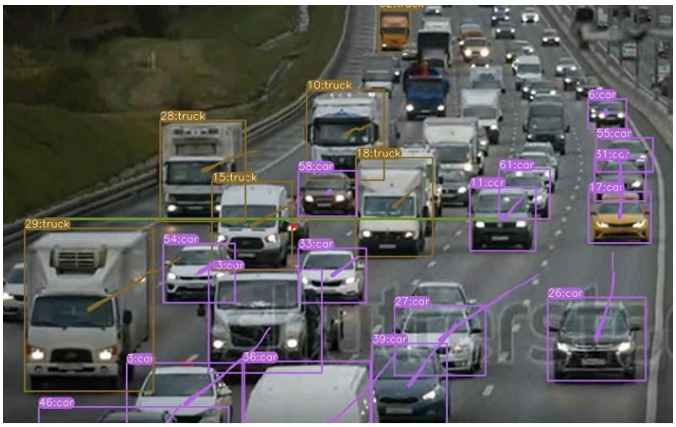


Fig. 5. Vehicle Detection in crowd and Assign Id

mean Average Precision (mAP) is an metric used for evaluating the machine learning model, which is the combination of recall and precision for object detection. For measuring mAP we need to review classification and localization. Classification is to identifying the category or class of each object within an image where the model assigns certain probability score to each detected objects giving its certainty about the class of that object. Localization involves predicting the position and drawing the boundary box around the object whenever an object of the relevant class which appears in the image.

In object detection, Intersection Over Union (IoU) is used to measure the overlap between the predicted boundary box (depicted in red) and the ground truth boundary box (shown in green) as illustrated in Figure 7. IoU is a crucial metric in this domain, and it serves as the basis for calculating Mean Average Precision (mAP), which evaluates model performance using IoU values at specific threshold levels. Here if the threshold value is set to 0.5 and result produced by the model is below 0.5 then it comes under false positive and above 0.5 as true positive

$$mAP = \frac{1}{N}\sum_{i=1}^{N} AP_i$$

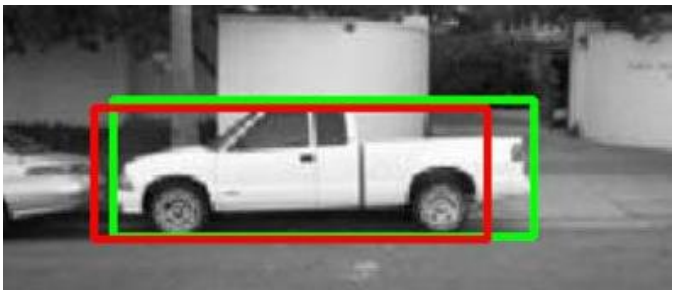


Fig. 6. Yolo IOU Detection

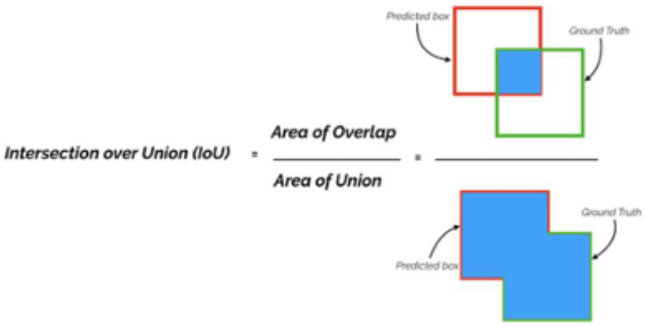


Fig. 7. IOU Definition

The mean Average Precision (mAP) is calculated by taking mean of the Average Precision (AP) across all classes and general IoU thresholds, depending on the various detection challenges encountered. The COCO dataset includes 1.5 million object instances across 80 classes and has its annotations in JSON files. For the COCO challenge, the IoU thresholds are set to range from 0.5 to 0.95 in increments of 0.05, resulting in a total of nine iterations. The computed APs at each threshold are then averaged to obtain the mAP.

TABLE II. Model mAP PERFORMANCE

Model	mAP %	Instances	Dataset
YOLO V8	53.9%	80	COCO
YOLO V7	47.2%	80	COCO
YOLO V5	43.2%	80	COCO
YOLO V4	40.2%	80	COCO

The performance of different YOLO models – YOLOv8, YOLOv7, YOLOv5 and YOLOv4 – was evaluated using the COCO dataset, which consists of 80 object classes. Each model was evaluated based on the mean precision (mAP) at the cutoff point above the Union threshold (IoU) of 0.5, which is a completely independent calculation for object detection, YOLOv8 demonstrates mAP@0.5 of 53.9 %, which is quite comparable in its ability to detect and classify objects across 80 classes in the COCO dataset with high accuracy. YOLOv7 achieves mAP@0.5 of 47.2 % and maintains strong performance in object identification and localization. YOLOv5 produces mAP@0.5 of 43.2 %, which indicates stable detection synthesis across categories in On the final dataset, YOLOv4 registers mAP@0.5 at 40.2%, indicating its ability to handle object detection tasks. The results highlight the strength of each model in object detection. Especially when compared to comprehensive datasets such as COCO, these mAP screens demonstrate the ability of each model to effectively generalize across different object classes. This makes it suitable for a variety of uses within data visualization. By testing the COCO validation dataset, the evaluation highlights the relevance of each model when detecting objects in different and complex environments. Each version of YOLO contributes to model understanding and performance. Detect and helps in selecting the most appropriate model for a specific job. This detailed evaluation serves to demonstrate the performance of the YOLO architecture in the face of challenges posed by large image datasets.

V. *CONCLUSIONS*

This project develops an intelligent traffic light control system using YOLOv8, an advanced object detection algorithm. To optimize traffic flow by dynamically adjusting signal timing according to vehicle density at time. Video capture analysis system Detects and counts the driver's direction in each image. To be able to adapt to changing traffic conditions and reduce overload. Especially during rush hour Contrast that with traditional rush hour signals. This system reduces bus travel time to a minimum. Reduce vehicle congestion and improve overall traffic efficiency YOLOv8's high accuracy and fast processing speed allow for almost instantaneous adjustment of traffic lights. This makes it suitable for real-time use. This project demonstrates the power of AI-powered traffic management solutions and demonstrates how deep learning and data visualization can meet urban challenges. Future iterations may include additional factors, such as changing the footprint. Pressure driving conditions or conditions of the carrier for more comprehensive adjustments This project reinforces the potential of totally intelligent systems to improve urban mobility. Reduce fuel use and support sustainable urban growth.

This system intelligent traffic light control system can be improved through the integration of advanced technologies such as AI-powered predictive analytics and vehicle-to-infrastructure (V2I) communication. These improvements will help the system detect Can only be caught by ambulance. Instead, the system will forward the best route based on current traffic patterns, road conditions and potential hazards. Therefore, the system will be able to dynamically adjust traffic lights to provide ambulances with the fastest possible route. It also reduces response time. Additionally, a GPS communication system will be installed for ambulances. which can directly interact with traffic lights This greatly improves the detection speed and response speed. This allows the system to predict the arrival of an ambulance during a crisis. and improve traffic spice processes. In addition to the ambulance the system can also be expanded to cover other pressurized vehicles, such as fire trucks and police vehicles, and can create a wider emergency network. This will help improve general mobility in the city. This will reduce delays. Reduces overloading in emergency situations and helps critical crew reach their destination as quickly and safely as possible. These innovations are an important step towards fully automated traffic control systems. This helps optimize traffic flow in emergency and normal situations in increasingly chaotic urban environments.

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